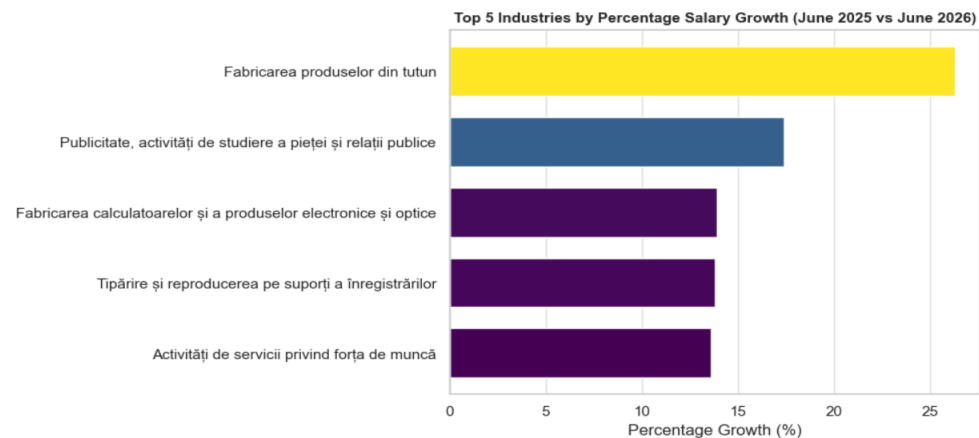
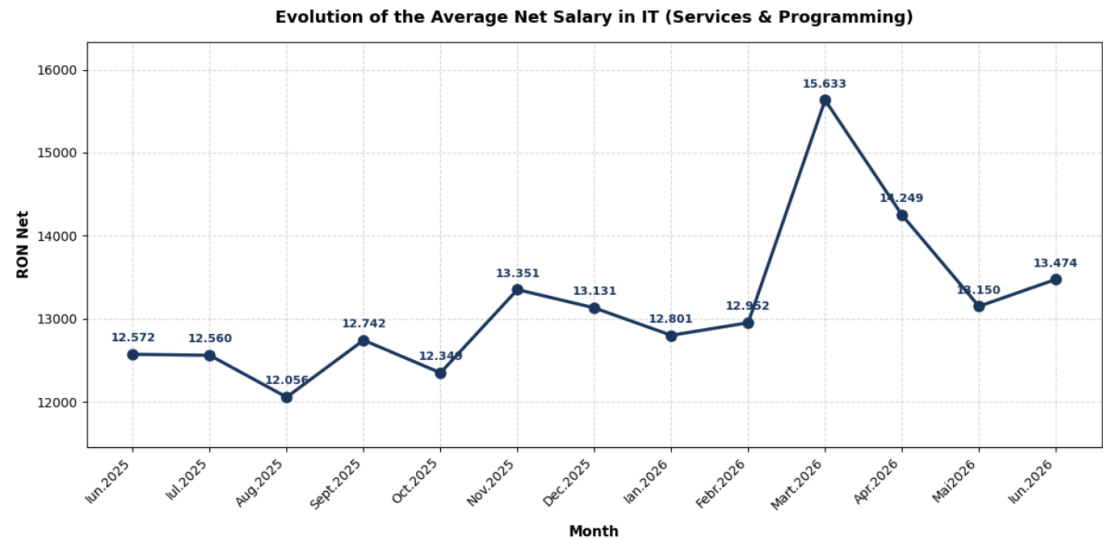


Net Salary Analysis

Analyzed official INS net salary dataset (2025–2026) using Python to track monthly salary evolution and sector disparities, visualizing monthly fluctuations and top industry performers with Matplotlib.

❖ Results

- Average Net Monthly Earnings: Reached 5,734 RON, recording an increase of 50 RON (+0.9%) compared to May 2026.
- Annual Growth Rate: Average net monthly earnings increased by +3.5% compared to June 2025.
- Highest Earnings by Sector: The highest net monthly salaries were recorded in **computer programming and information technology** consultancy activities (13,474 RON) and publishing activities (11,999 RON).
- Lowest Earnings by Sector: The lowest net monthly salaries were recorded in other service activities (2,897 RON) and fishing and aquaculture (2,912 RON).
- The IT sector net average salary grew overall by +5.89% annually (reaching 11,800 RON in June 2026), displaying high monthly volatility with a lowest point in August 2025 (10,638 RON) and a major peak in March 2026 (13,352 RON).



❖ Methodology

- Data manipulation on Excel files, computed annual and monthly percentage changes, created trend lines and comparative bar charts.
- Compute Average net salary in June 2026 compared to May 2026 by formula: $iun_mai = ((iun_2026 - mai_2026)/mai_2026)*100$
- Compute Average net salary in June 2026 compared to June 2025 by formula: $iun2026_iun2025 = ((iun_2026 - iun_2025)/iun_2025)*100$
- Sort descending and ascending by the 'salary' column to find the highest and lowest earnings by sector.
- Select the industry by CAEN code to determine the sector with the highest salary.
- Compute the percentage growth to identify the Top 5 Industries by Percentage Salary Growth (June 2025 vs June 2026) - $((iun_2026 - iun_2025)/iun_2025)*100$

❖ Tools used

- Python (Pandas, Matplotlib, Seaborn), Microsoft Excel.

❖ Data

- Data Source: National Institute of Statistics (INS Romania)
- Dataset: Official monthly average net earnings by economic activity (CAEN Rev. 2 sections and divisions / 80+ sectors)
- Time Period: June 2025 – June 2026

❖ Problem

- Tracking national wage dynamics and identifying cross-sector salary disparities and volatility.

Python Script

Below is the full Python script utilized for data loading, wrangling, metric aggregation, and generating visualizations.

```
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

# =====
# LOAD DATA
# =====
file_path = (
    "C:/Users/AnaTiru/Desktop/Data_Analyst/Curs_data_analyst_SKILLAB/Lectia 20"
    "- 12.08.2026 Interviu/net_salary_ins.xlsx"
)
df = pd.read_excel(file_path)
# =====
# 1. Average net salary in June 2026 compared to May 2026
```

```

# =====
iun_2026 = df["Iun.2026"].iloc[0]
mai_2026 = df["Mai2026"].iloc[0]
pct_iun_mai = round(((iun_2026 - mai_2026) / mai_2026) * 100, 1)
print(
    f"Average net salary in June 2026 increased by +{pct_iun_mai}% compared to"
    " May 2026."
)
# =====
# 2. Average net salary in June 2026 compared to June 2025
# =====
iun_2026 = df["Iun.2026"].iloc[0]
iun_2025 = df["Iun.2025"].iloc[0]
pct_iun_iun = round(((iun_2026 - iun_2025) / iun_2025) * 100, 1)

print(
    f"Average net salary in June 2026 increased by +{pct_iun_iun}% compared to"
    " June 2025."
)
# =====
# 3. Highest Earnings by Sector
# =====
order_df_desc = df.sort_values(by="Iun.2026", ascending=False)
highest = order_df_desc.iloc[0:2][["Industry", "Iun.2026"]]
print(
    f"Highest Earnings by Sector: {highest['Industry'].iloc[0]} with"
    f" {highest['Iun.2026'].iloc[0]} RON and {highest['Industry'].iloc[1]} with"
    f" {highest['Iun.2026'].iloc[1]} RON"
)
# =====
# 4. Lowest Earnings by Sector
# =====
order_df_asc = df.sort_values(by="Iun.2026", ascending=True)
lowest = order_df_asc.iloc[0:2][["Industry", "Iun.2026"]]
print(
    f"Lowest Earnings by Sector: {lowest['Industry'].iloc[0]} with"
    f" {lowest['Iun.2026'].iloc[0]} RON and {lowest['Industry'].iloc[1]} with"
    f" {lowest['Iun.2026'].iloc[1]} RON"
)
# =====
# 5. Monthly Evolution of the Net Average Salary in IT (Services & Programming)
# =====
it_salary = df[df["Cod CAEN Rev.3"] == 62]
salaries = it_salary.iloc[0, 2:15].values.astype(int)
months = df.columns[2:15].tolist()
plt.figure(figsize=(12, 6))
plt.plot(
    months, salaries, marker="o", color="#1A365D", linewidth=2.5, markersize=8

```

```

)
min_sal = min(salaries)
max_sal = max(salaries)
plt.ylim(min_sal - 600, max_sal + 700)

plt.xlabel("Month", fontsize=11, fontweight="bold", labelpad=10)
plt.ylabel("RON Net", fontsize=11, fontweight="bold")
plt.title(
    "Evolution of the Average Net Salary in IT (Services & Programming)",
    fontsize=13,
    fontweight="bold",
    pad=15,
)
plt.xticks(rotation=45, ha="right", fontsize=10)
plt.yticks(fontsize=10)
plt.grid(True, linestyle="--", alpha=0.5)

for i, txt in enumerate(salaries):
    plt.annotate(
        f"{txt:,.} ".replace(",", ". "),
        (months[i], salaries[i] + 120),
        ha="center",
        va="bottom",
        fontweight="bold",
        fontsize=9,
        color="#1A365D",
    )

plt.tight_layout()
plt.show()

# =====
# 6. Top 5 Industries by Percentage Salary Growth (June 2025 vs June 2026)
# =====
df["procent"] = round(
    ((df["Iun.2026"] - df["Iun.2025"]) / df["Iun.2025"]) * 100, 1
)
df_proc = df.sort_values(by="procent", ascending=False)
top_5 = df_proc.iloc[0:5]

date = pd.DataFrame({
    "Domeniu": [
        "Fabricarea produselor din tutun",
        "Publicitate, activități de studiere a pieței și relații publice",
        "Fabricarea calculatoarelor și a produselor electronice și optice",
        "Tipărire și reproducerea pe suporti a înregistrărilor",
        "Activități de servicii privind forța de muncă",
    ],

```

```

    "Crestere": [26.3, 17.4, 13.9, 13.8, 13.6],
})

date_sorted = date.sort_values(by="Crestere", ascending=True)

plt.figure(figsize=(10, 5))
sns.set_theme(style="whitegrid")

norm = plt.Normalize(
    date_sorted["Crestere"].min(), date_sorted["Crestere"].max()
)
colors = plt.cm.viridis(norm(date_sorted["Crestere"]))

plt.barh(
    date_sorted["Domeniu"], date_sorted["Crestere"], color=colors, height=0.7
)
plt.title(
    "Top 5 Industries by Percentage Salary Growth (June 2025 vs June 2026)",
    fontsize=11,
    fontweight="bold",
)
plt.xlabel("Percentage Growth (%)")
plt.gca().grid(True, axis="x", color="#E0E0E0")
plt.gca().grid(False, axis="y")

plt.tight_layout()
plt.show()

```